

AlphaAI Fund Platform

Decentralized AI-Powered Hedge Fund

Technical Documentation & Platform Overview

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Status: Production Ready (Testnet)
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1. Executive Summary

AlphaAI is a decentralized AI-powered hedge fund platform that enables investors to deposit capital into a vault managed by autonomous AI trading agents. The platform features advanced simulation capabilities, real-time market data integration, and comprehensive risk management systems.

Key Highlights

Feature	Status	Description
Live Price Feed	■ Active	Real-time prices from Kraken API
MetaMask Integration	■ Active	Web3 wallet connection for Sepolia
AI Trading Agents	■ Active	4 autonomous trading strategies
Research Engine	■ Active	500x accelerated backtesting
Smart Contract	■ Ready	AlphaAIManager.sol for Sepolia
Walk-Forward Testing	■ Active	Strategy robustness validation
Risk Management	■ Active	Auto-stop, drawdown limits
Event Agents	■ Active	3 event-driven monitors

2. Platform Architecture

Technology Stack

Component	Technology	Purpose
Frontend	React + Shadcn/UI	User interface
Backend	FastAPI (Python)	API server
Database	MongoDB	Data persistence
Blockchain	Ethereum (Sepolia)	Smart contracts
Wallet	MetaMask + ethers.js	Web3 integration
Price Data	Kraken API	Live market prices
AI Analysis	OpenAI GPT-5.2	Market analysis

Directory Structure

```
/app/  
├── backend/  
│   ├── server.py (Main FastAPI application)  
│   ├── contracts/AlphaAIManager.sol  
│   ├── data/ (Historical price CSV files)  
│   ├── reports/ (Generated investor reports)  
├── frontend/  
│   ├── src/App.js (Main React application)  
│   ├── components/ui/ (Shadcn components)  
└── memory/PRD.md (Product requirements)
```

3. AI Trading Agents

The platform employs four specialized AI trading agents, each allocated 25% of the fund's capital. These agents operate autonomously using different trading strategies to maximize risk-adjusted returns.

Agent Name	Strategy	Allocation	Description
Arbitrage Agent	Mean Reversion	25%	Trades against large price moves, capitalizing on market overreactions
Momentum Agent	Trend Following	25%	Follows established trends with enhanced position sizing
Funding Rate Agent	Volatility Breakout	25%	Trades on volatility spikes and market momentum
AI Research Lab	Adaptive	25%	Auto-generates and deploys new strategies based on market conditions

4. Smart Contract Integration

AlphaAIManager.sol

The AlphaAIManager smart contract is deployed on the Ethereum Sepolia testnet and handles:

- Investor deposits and withdrawals
- Strategy fund allocation
- On-chain balance tracking
- Emergency withdrawal functionality (owner only)

Function	Type	Description
deposit()	payable	Deposit ETH into the fund
withdraw(uint256)	nonpayable	Withdraw specified amount
addStrategy(string)	onlyOwner	Add new trading strategy
allocateToStrategy(uint256, uint256)	onlyOwner	Allocate funds to strategy
getInvestorBalance(address)	view	Get investor balance
getStrategy(uint256)	view	Get strategy details
emergencyWithdraw()	onlyOwner	Emergency fund withdrawal

5. Research Engine

The AlphaAI Research Engine provides institutional-grade backtesting and strategy validation capabilities with 500x time acceleration.

Features

Feature	Configuration	Output
Accelerated Simulation	500x speed, 6 months	Equity curve, metrics, trade history
Walk-Forward Testing	90d train / 30d test	Robustness score, window results
Historical Data	BTC/USD, ETH/USD CSV	Price-driven simulation
Performance Metrics	14 key metrics	Sharpe, Sortino, Max DD, etc.
Strategy Contribution	Per-agent breakdown	Individual P&L attribution
Investor Reports	PDF/JSON format	Monthly performance reports

Performance Metrics Tracked

- Total Return & Annualized Return
- Sharpe Ratio & Sortino Ratio
- Maximum Drawdown
- Win Rate & Profit Factor
- Trade Frequency
- Average Trade Return
- Best/Worst Trade
- Strategy Contribution %

6. Risk Management

The platform includes comprehensive risk management controls to protect investor capital.

Control	Threshold	Action
Max Drawdown	5%	Auto-stop all trading
Daily Loss Limit	3%	Reduce position sizes by 50%
Position Size	10-15%	Per-trade allocation limit
Stop Loss	5%	Per-agent stop loss
Risk Alerts	2% drawdown	Send alerts to all agents

Stress Test Scenarios

Scenario	Parameters	Purpose
BTC 30% Drop	30% drop over 24 hours	Test market crash resilience
ETH Flash Crash	50% drop over 12 hours	Test flash crash handling
Market Panic	40% drop over 6 hours	Test rapid selloff response
Liquidity Crisis	25% drop over 48 hours	Test extended stress periods

7. Event-Driven Agent System

Three specialized agents monitor smart contract events and execute automated responses.

Agent	Type	Events Monitored	Actions
Investor Monitor	Watcher	InvestorDeposited, InvestorWithdrawn	Update balances dashboard
Strategy Allocator	Execution	InvestorDeposited	Auto-allocate 25% to top strategy
Dashboard Updater	Analytics	StrategyAllocated, StrategyDeallocated	Refresh allocation dashboard

8. Live Market Data Integration

Real-time cryptocurrency prices are fetched from the Kraken API with 30-second refresh intervals.

Asset	Symbol	Data Points
Bitcoin	BTC/USD	Price, 24h Change, Volume
Ethereum	ETH/USD	Price, 24h Change, Volume
Solana	SOL/USD	Price, 24h Change, Volume
Avalanche	AVAX/USD	Price, 24h Change, Volume
Polygon	MATIC/USD	Price, 24h Change, Volume
Chainlink	LINK/USD	Price, 24h Change, Volume
Uniswap	UNI/USD	Price, 24h Change, Volume
Aave	AAVE/USD	Price, 24h Change, Volume

10. API Reference

Endpoint	Method	Description
/api/market/live-prices	GET	Get live crypto prices
/api/simulation/start	POST	Start simulation
/api/simulation/run-accelerated	POST	Run 500x simulation
/api/simulation/stress-test	POST	Run stress test
/api/research/run-simulation	POST	Run research backtest
/api/research/walk-forward-test	POST	Run walk-forward validation
/api/research/generate-investor-report	POST	Generate PDF report
/api/contract/info	GET	Get contract status
/api/contract/register	POST	Register deployed contract
/api/agents/event-agents	GET	Get event agents
/api/dashboards/investor-balances	GET	Get investor balances
/api/dashboards/strategy-allocation	GET	Get allocations
/api/fund/stats	GET	Get fund statistics
/api/lab/strategies	GET	Get all strategies
/api/risk/config	GET/PUT	Risk configuration

11. Security & Compliance

Security Measures

- MetaMask wallet integration for secure authentication
- Smart contract owner-only functions for critical operations
- Emergency withdrawal capability for fund protection
- Rate limiting on API endpoints
- Input validation on all user inputs

Pre-Mainnet Requirements

- Smart contract security audit (recommended: OpenZeppelin, Trail of Bits)
- Penetration testing of web application
- JWT user authentication implementation
- Legal and regulatory compliance review
- Insurance coverage consideration

12. Development Roadmap

Phase	Status	Features
Phase 1: MVP Core	■ Complete	Basic trading agents, dashboard, marketplace
Phase 2: Advanced AI	■ Complete	Strategy lab, risk engine, capital allocation
Phase 3: Simulation	■ Complete	Paper trading, trade cycles, logging
Phase 4: Reports	■ Complete	Daily/weekly reports, mode switching
Phase 5: Enhanced Sim	■ Complete	500x acceleration, stress testing
Phase 6: Smart Contract	■ Complete	Solidity contract, MetaMask integration
Phase 7: Research Engine	■ Complete	Walk-forward validation, CSV data
Phase 8: Testnet Deploy	■ Pending	Deploy to Sepolia, live testing
Phase 9: Security Audit	■ Pending	Third-party audit, penetration testing
Phase 10: Mainnet	■ Pending	Production deployment, real trading

Production Readiness: 90%

